Next-generation Bayesian local robustness

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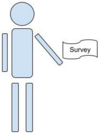
Berkeley Neyman Seminar
September 2026

The basic problem

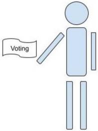
We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to "do you support candidate Z")

We want the average response in a target population, in which we observe only covariates.



Observe (x_i, y_i) for $i = 1, \ldots, N_S$
(This is the "Data", $\mathcal{D}$)



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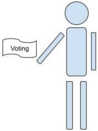

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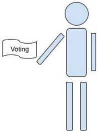

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This talk will study “Multilevel regression and poststratification” estimators:

- Set $\mathbb{E}\left[y|x, \theta\right] = m(\theta^\intercal x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathcal{D})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[y|x_j\right]$.

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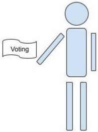

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Problem: MCMC is time consuming and the regression model is surely misspecified.
Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

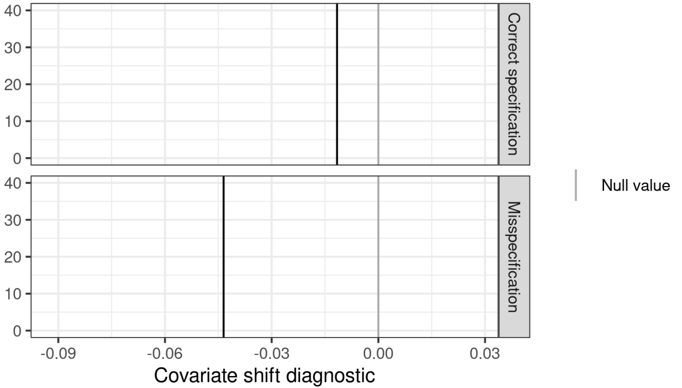

Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecied model is more extreme.

Problem: What is "large enough" to be real evidence of misspecification?
Possible answer: Frequentist confidence intervals via the parametric bootstrap.

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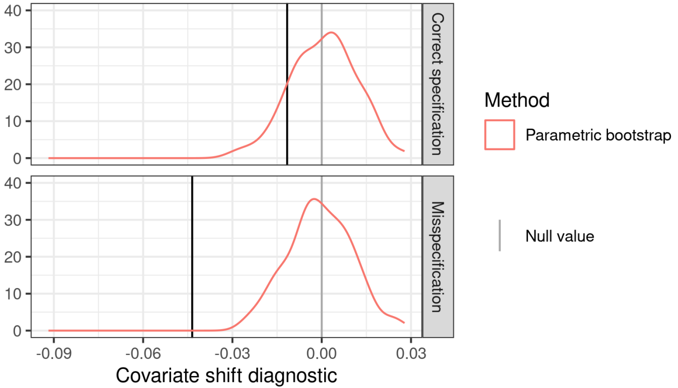

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The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.
Possible answer: Infintesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

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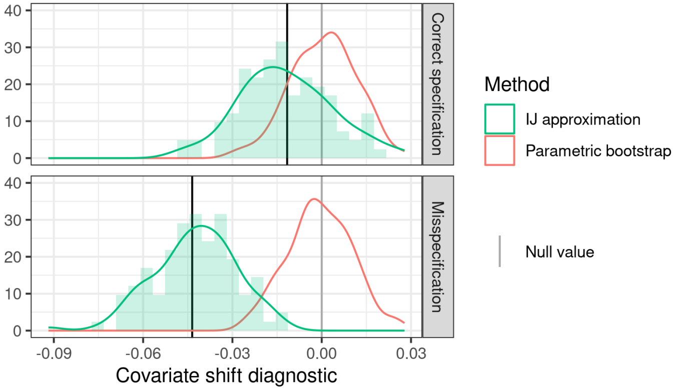

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IJ confidence intervals are fast and often work well for posterior means.
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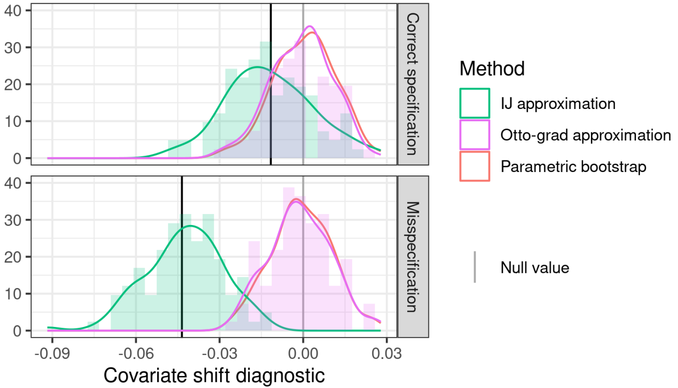

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Answer: Our flow-based local sensitivity measures ("Otto grad") produce good approximations to the bootstrap at negligible computational expense.

Outline

- Bayesian local regression specification checks
  - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the inifinitesimal jackknife
  - (With Tamara Broderick)
- Flow-based Bayesian local robustness: “Otto grad”
  - (With Lucas Schwengber)

TODO: put in pictures

Bayesian local regression specification checks

Local specification checks

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\intercal x_j) \right]}_{\substack{\text{Want to check specification} \\ \text{without re–running MCMC}}} .$$

Some existing local approaches:

- Specify an alternative model, maybe test with Bayes factor [Di Noia et al., Cabral et al.]
  - Bayes factors can depend strongly on priors (Lindley's paradox)
  - Articulating a family of models is inconvenient
  - Doesn't take the target into account
- Use out-of-sample checking [Vehtari et al., Kuh et al.]
  - The task is not to predict on a new IID survey datapoint

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The model is definitely misspecified, we want to check whether it's *importantly* misspecified.

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Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y.
Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x)\mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x))\, \mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial\mu(\mathcal{P}_\epsilon)/\partial\epsilon$
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Lots of similar ideas in causal inference, kernel learning, transfer learning [Bühlmann, Rojas-Carulla et al., Muandet et al.]. This is just a local Bayesian diagnostics version.

Simple visual example

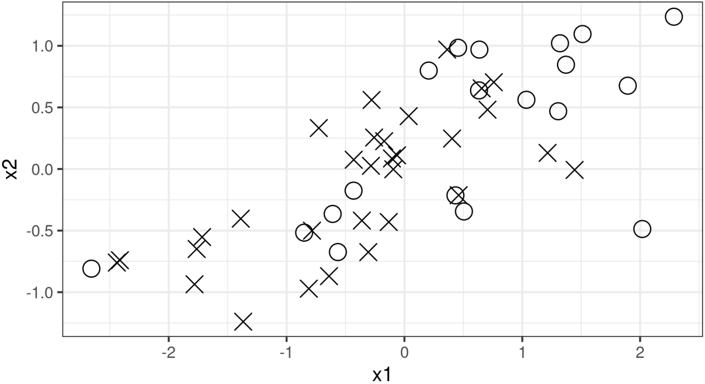

Figure 2: A covariate shift diagnostic

Consider logistic regression with two regressors, x_1 and x_2.

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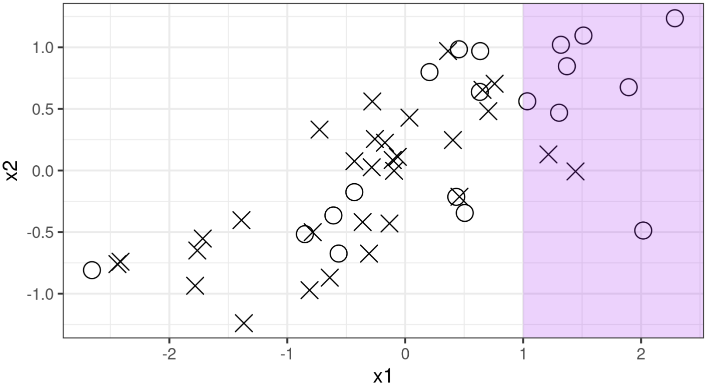

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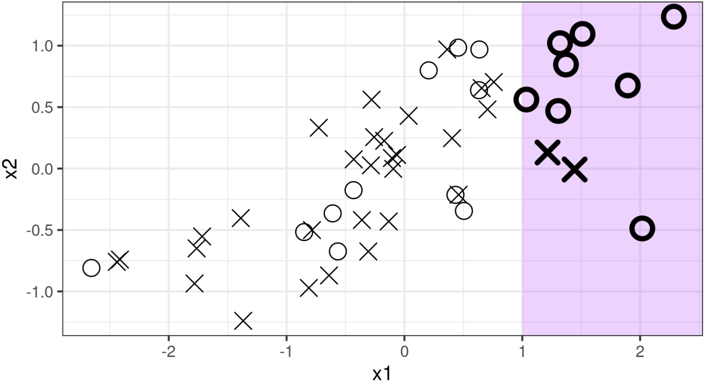

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With enough data, a regression that omits x_1 will change its coefficient of x_2 under re-weighting, whereas a correctly specified regression will not need to.

Bayesian sensitivity analysis

Let the regression log likelihood be $\ell(y|x,\theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i,\theta)\right) \pi(\theta) \text{ and } \mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i)\ell(y_i|x_i,\theta)\right)$$

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Theorem (Diaconis and Freedman, Gustafson [1996], Giordano et al. [2018]):
Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$ and a function $\phi(\theta)$. By interchange of integration and differentiation,

$$\left.\frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi(\theta), h(\theta)\right) .$$

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Our diagnostic is then given by

$$\Delta(\mathcal{D}) := \left.\frac{\partial \hat{\mu}}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta|\mathcal{D})}\left(\underbrace{\frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^{\intercal} x_j)}_{\text{Estimand}}, \underbrace{\sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i,\theta)}_{\text{Log model perturbation}}\right)$$

We expect $\Delta(w,\mathcal{D})$ to be *close* to zero when the model is correctly specified.

Bootstrapping Bayesian sensitivity analysis

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But on any actual dataset, $\Delta(w, \mathcal{D})$ will typically not be precisely zero, even under correct specification.

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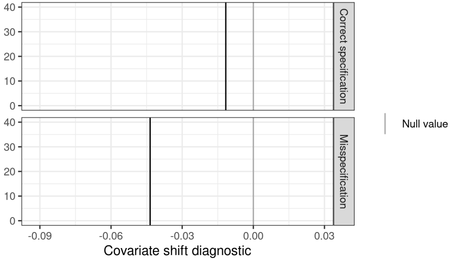

Figure 3: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

$$\text{Model 0 (correct)} : \texttt{y} \sim \text{Logistic}(\texttt{1 + X1 + X2 + X3 + X4 + X5})$$

$$\text{Model 1 (misspecified)} : \texttt{y} \sim \text{Logistic}(\texttt{1 + X3 + X4 + X5})$$

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To assess "closeness to zero," compare with $\Delta(\mathcal{D}^*)$, where $\mathcal{D}^*$ is a bootstrap sample.

- Compute $\Delta(\mathcal{D})$ on the original dataset
- For $b = 1, \ldots, 100$:
  1. Sample $\mathcal{D}_b^* := (x_1^*, y_1^*), \ldots (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
  2. Run MCMC to estimate $\mathcal{P}(\theta|\mathcal{D}_b^*)$ **(time consuming!)**
  3. Use the draws to estimate $\Delta(\mathcal{D}_b^*)$
- Compare Δ with distribution of $\Delta(\mathcal{D}_b^*)$.

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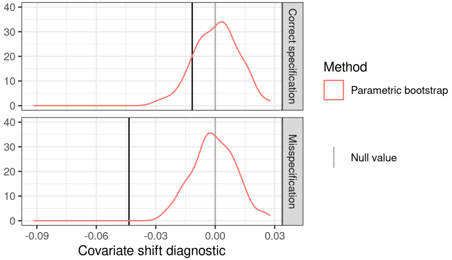

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Frequentist variance with the infinitesimal jackknife

Why frequentist variance?

For the remainder of the talk, focus on *fast approximations* to the *frequentist* sampling distribution under $x_i, y_i \overset{iid}{\sim} \mathcal{P}_S(x, y)$ of Bayesian posterior quantities like our diagnostic.

We will form *local approximations* computable only from *the original posterior*.

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We will form *local approximations* computable only from *the original posterior*.

No need to commit to our covariate shift diagnostic to care!

- Frequentist variablity of *any* model checking diagnostic is interesting
- Frequentist variablity remains meaningful under model misspecification [Huggins and Miller, 2023, Wang et al., 2026, Luo and Ji, 2026, Ji et al., 2025]

We will compare with computationally expensive full bootstrap.

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x,\theta)$, and fix $w(x)$. Then define

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We can form a series approximation:

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$$=$$

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Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x,\theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i,\theta)\right)\pi(\theta) \text{ and } \mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i)\ell(y_i|x_i,\theta)\right)$$

If we let $w(x_i)$ denote a *bootstrap weight*, then $\mathcal{P}^{\epsilon}$ at $\epsilon = 1$ is the bootstrapped posterior!

We can form a series approximation:

$$\begin{aligned}
\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D}^*)}\left[\phi(\theta)\right] &\approx \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] + \left.\frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} (1-0) \\
&= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] + \sum_{i=1}^{N_S} \underbrace{\left.\frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[\phi(\theta)\right]}{\partial w_i}\right|_{w_i=0}}_{\text{"Empirical influence function" } \frac{\psi_i}{N_S}} w_i \\
&= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] + \frac{1}{N_S}\sum_{i=1}^{N_S} \psi_i w_i
\end{aligned}$$

If this series is accurate, then we can compute ψ_i using the original posterior and quickly approximate bootstrap draws.

Bayesian von–Mises Expansion

Define the "generalized posterior" functional [Giordano and Broderick, 2026]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dx\,dy\right)\pi(\theta)d\theta}{\int \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dx\,dy\right)\pi(\theta)d\theta}.$$

Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] = T(\hat{\mathcal{P}}_S, N_S)$

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To derive frequentist variance estimates, we study the *von Mises expansion* [von Mises, 1947]. Let $\tilde{\mathcal{P}}^t = t\hat{\mathcal{P}}_S + (1-t)\mathcal{P}_S$. Then form the series expansion in t:

$$\begin{aligned}
\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) &= \sqrt{N_S}\left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S)\right) \\
&= \sqrt{N_S}\left.\frac{\partial T(\tilde{\mathcal{P}}^t, N_S)}{\partial t}\right|_{t=0} (1-0) + \mathcal{E}(\tilde{t}) \quad (\tilde{t} \in [0,1]) \\
&= \underbrace{\sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}).
\end{aligned}$$

Bayesian von–Mises Expansion Results

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) = \sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

Theorem 3 [Giordano and Broderick, 2026] (sketch):
Under Bernstein-von Mises-like conditions, $\sup_{\tilde{t}\in[0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{p} 0$.

Corollary [von Mises, 1947]: Let $\frac{1}{N_S}\sum_{i=1}^{N_S}(\psi_i - \overline{\psi})^2 \xrightarrow{p} V$. By the CLT applied to ψ_i,

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) \xrightarrow{d} \mathcal{N}\left(0, V\right).$$

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To compute and study the parametric bootstrap instead is a simple change:

$$\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon\left(\ell(y_i^*|x_i,\theta) - \ell(y_i|x_i,\theta)\right)\right).$$

(See, e.g., Giordano et al. [2026] Theorem 4.1).

Bayesian von–Mises Expansion Results

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) = \sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

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(See, e.g., Giordano et al. [2026] Theorem 4.1).

Note: The von-Mises expansion is stronger than necessary for consistency [Giordano and Broderick, 2026]. But it can be used to prove *lower bounds* on $\mathcal{E}(\tilde{t})$ and so *inconsistency* when no BVM theorem holds.

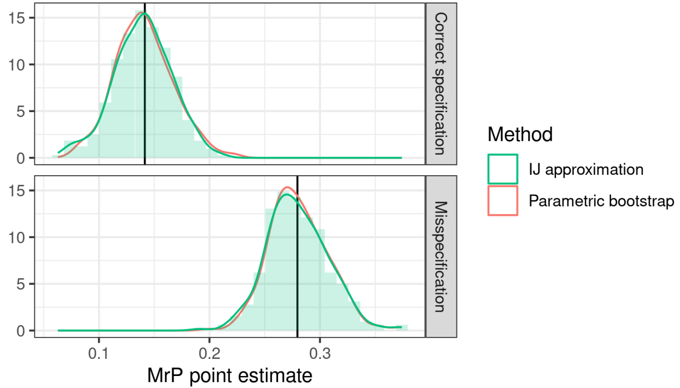

Figure 4: MrP Posterior Mean

The IJ does great for the MrP point estimate. See also Luo and Ji [2026], Ji et al. [2025].

Data re-weighting.

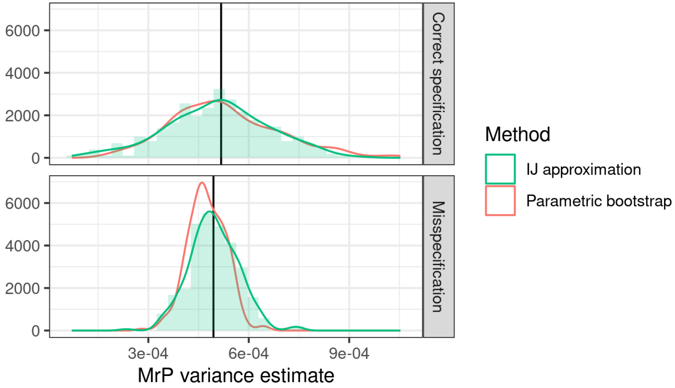

Figure 4: MrP Posterior Variance

The IJ does pretty well for the MrP posterior variance.

Data re-weighting.

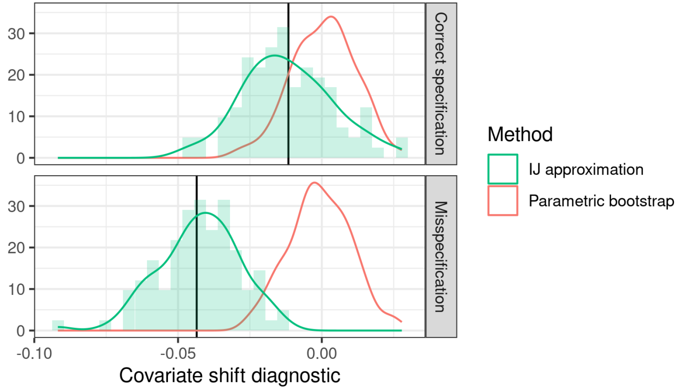

Figure 4: Covariate Shift Diagnostic

Need something better than the IJ for noisier / more complex posterior quantities!

Local robustness with flows

Why flows for sensitivity analysis?

Current Bayesian local robustness is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[\phi(\theta)\right] \approx \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta)\right] + \left.\frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} (\epsilon - 0),$$

where the derivative is estimated — with noise — from MCMC draws [Gustafson, 1996, Giordano et al., 2018, Cabral et al., Di Noia et al., Giordano et al., 2026].

This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantiles are hard to linearize
- Bayesians like to work with draws, not analytic expectations (CITE)

Idea: Instead of linearizing expectations, learn a flow from $\mathcal{P}$ to $\mathcal{P}^{\epsilon}$.

There's a large literature on learning flows between distant distributions. (CITE)
Our problem is easier:

- Only need "local" flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Flow

Consider $\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

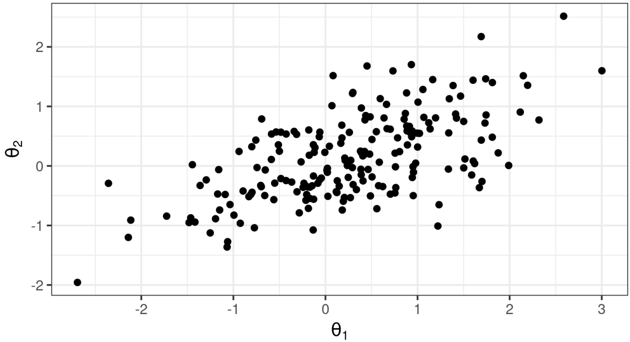

Figure 5: Draws from a posterior

Flow

Consider $\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

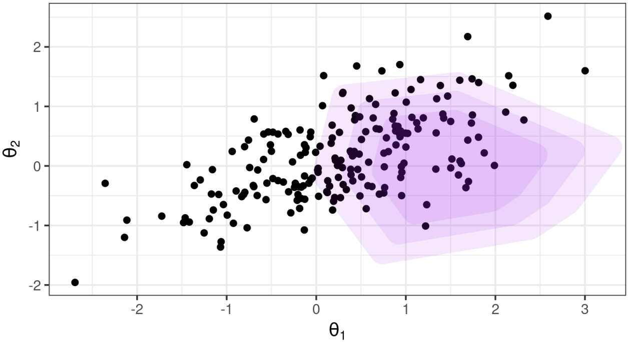

Figure 5: A posterior perturbation

Imagine the density $\mathcal{P}(\theta)$ coming out of the page. The perturbation $h(\theta)$ is defined "vertically."

Flow

Consider $\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

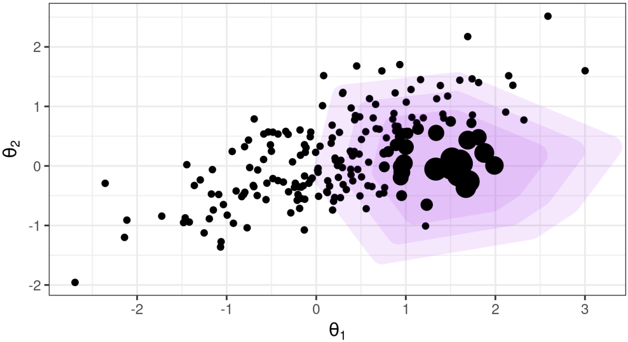

Figure 5: A posterior perturbation with weights

Plug-in estimates for the covariance are equivalent to producing the perturbation by weighting (Appendix B Giordano et al. [2018])

Flow

Consider $\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

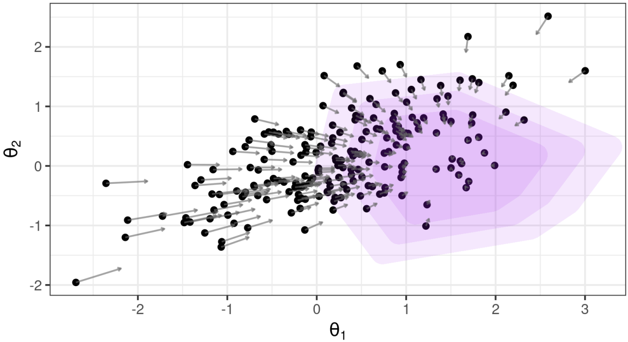

Figure 5: A posterior perturbation with a flow

We will try, instead, try to effect the perturbation with a flow $\theta \mapsto f^{\epsilon}(\theta)$

Flow

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ, $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

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Suppose there is a scalar-valued potential function $v(\theta)$ such that

$$f^\epsilon(\theta) = \theta + \epsilon \nabla v(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \to 0. \text{ Then:}$$

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(\star) && \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi(\theta), h(\theta)\right) &= \mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla \phi(\theta)^\intercal \nabla v(\theta)\right] && \text{(Connecting } h \text{ and } v\text{)}
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\end{aligned}$$

Theorem (Schewengber 2026):
Under mild regularity conditions there exists a solution v to the differential equation

$$\mathcal{L}_{\mathcal{P}}(v) := -\nabla v^\intercal \nabla \log \mathcal{P} - \Delta v = h$$

such that $\|\nabla v\|_{L^2(\mathcal{P})} < \infty$ and the flow $\theta_\epsilon = \theta + \epsilon\nabla v(\theta)$ satsifes Equation $(\star)$.

Note that $\mathcal{L}_{\mathcal{P}}$ is well-known: it's the drift operator, generator of Langevin diffusion, and the trace of the Stein operator.[Ambrosio et al., 2021, Song et al., 2021, Ross, 2011]

How can we find $v(\theta)$ in practice?

Solving PDEs is hard. How can we easily find $v(\theta)$ in practice?

$$(\star) \qquad \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi(\theta), h(\theta)\right) = \mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla\phi(\theta)^\intercal \nabla v(\theta)\right] \qquad \text{(Identity connecting } h \text{ and } v\text{)}$$

Basis expansion approach (Otto's calculus [Otto, 2001]):

- Specify a family of real-valued test functions $\phi_n(\theta)$ for $n \in [N]$
- Specify a family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

Let $v(\theta; \beta) := \sum_{m=1}^{M} \beta_m b_m(\theta)$ and plug into Equation $(\star)$.

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Stacking in matrices:

$$\underbrace{\begin{bmatrix} \vdots \\ \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi_n(\theta), h(\theta)\right) \\ \vdots \end{bmatrix}}_{N \times 1 \text{ matrix}} \underbrace{\begin{bmatrix} & \vdots & \\ \dots & \mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla\phi_n(\theta)^\intercal \nabla b_m(\theta)\right] & \dots \\ & \vdots & \end{bmatrix}}_{N \times M \text{ matrix}} \begin{bmatrix} \vdots \\ \beta_m \\ \vdots \end{bmatrix}$$

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This gives N linear equations in M unknowns, and the matrices can be estimated with MCMC.

Theorem (Schewengber 2026): When $\mathcal{P}(\theta)$ is Gaussian, taking $N = M$ and $\phi_n(\theta) = b_n(\theta)$ to be the Hermite polynomials give the best finite-rank approximation to $v(\theta)$ in $L_2(\mathcal{P})$.

An elegant flow

Suppose we believe that $\mathcal{P}(\theta|\mathcal{D})$ is approximately Gaussian.

Using the 1^{st} order Hermite polynomials $\phi_n(\theta) = b_n(\theta) = \theta_m$, the estimated flow is given by

$$\theta_\epsilon = \theta + \epsilon \underbrace{\mathrm{Cov}_{\mathcal{P}(\theta|\mathcal{D})}\left(\theta, h(\theta)\right)}_{\frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon}\Big|_{\epsilon=0}}$$

This just moves all points in the direction of the mean shift!

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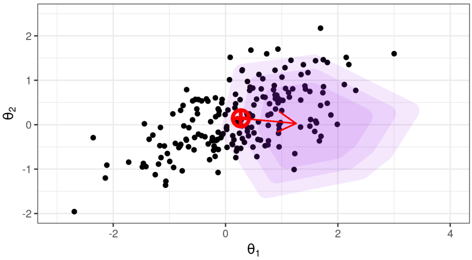

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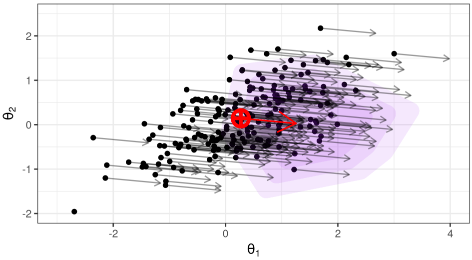

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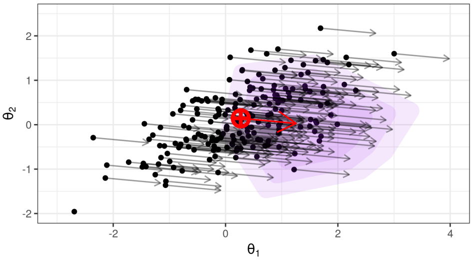

Suppose we have $\theta_m \sim \mathcal{P}(\theta|\mathcal{D})$, and bootstrap weights w_i.

Set $\theta_m^* = \theta_m + \frac{1}{N_S}\sum_{i=1}^{N_S} w_i \psi_i$ and compute any posterior quantities from these samples.

Data re-weighting.

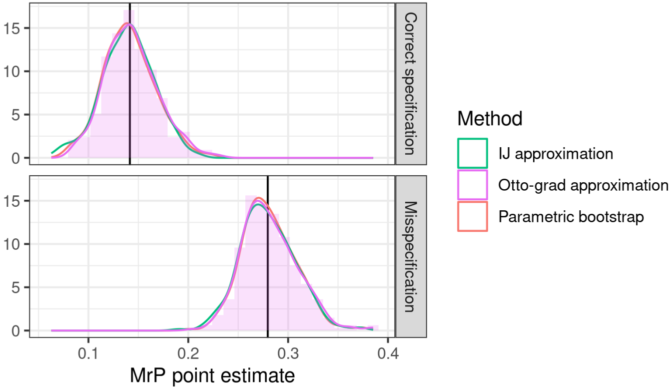

Figure 6: MrP Posterior Mean

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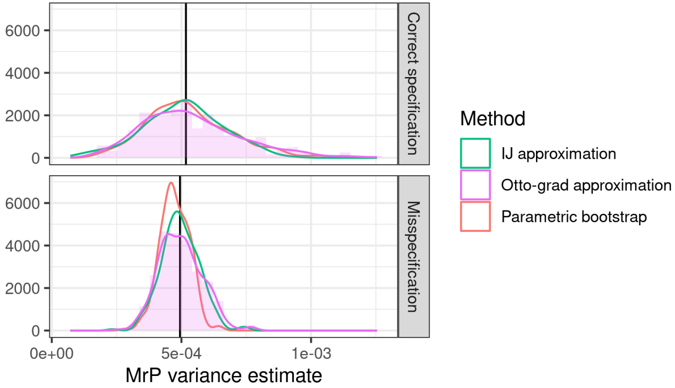

Figure 6: MrP Posterior Variance

Data re-weighting.

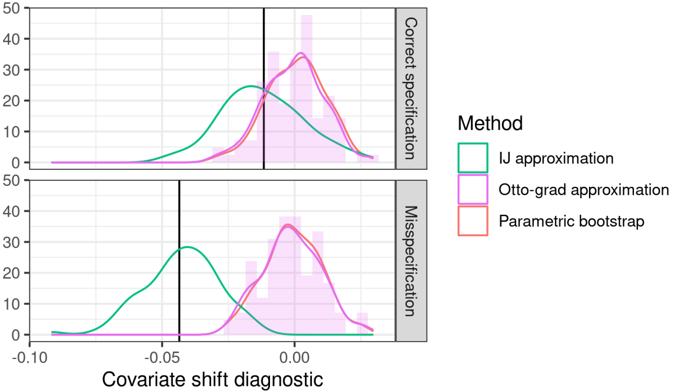

Figure 6: Covariate Shift Diagnostic

Understanding the gap between IJ and Otto is ongoing work!

Conclusion

- Bayesian local regression specification checks
  - Use invariance to covariate shift as a diagnostic for Bayesian hierarchical models
- Frequentist variability of Bayesian posteriors with the inifinitesimal jackknife
  - Use sensitivity of mean parameters to speed up bootstrapping MCMC
- Flow-based Bayesian local robustness: “Otto grad”
  - Use easy-to-compute approximate flows to speed up bootstrapping MCMC

Luigi Ambrosio, Elia Brué, Daniele Semola, et al. *Lectures on optimal transport*, volume 130. Springer Cham, 2021.

Peter Bühlmann. Invariance, Causality and Robustness: 2018 Neyman Lecture. 35(3):404–426. ISSN 0883-4237. URL `https://www.jstor.org/stable/26997912`.

Rafael Cabral, David Bolin, and Håvard Rue. Robustness, model checking, and hierarchical models. 87(3):632–652. ISSN 1369-7412, 1467-9868. doi: 10.1093/jrsssb/qkae107. URL `https://academic.oup.com/jrsssb/article/87/3/632/7914952`.

Antonio Di Noia, Fabrizio Ruggeri, and Antonietta Mira. Likelihood Distortion and Bayesian Local Robustness. 20(4). ISSN 1936-0975. doi: 10.1214/25-BA1526. URL `https://projecteuclid.org/journals/bayesian-analysis/volume-20/issue-4/Likelihood-Distortion-and-Bayesian-Local-Robustness/10.1214/25-BA1526.full`.

P. Diaconis and D. Freedman. On the consistency of Bayes estimates. pages 1–26.

R. Giordano and T. Broderick. The Bayesian infinitesimal jackknife for variance. *arXiv preprint arXiv:2305.06466*, 2026.

R. Giordano, T. Broderick, and M. Jordan. Covariances, robustness and variational Bayes. *The Journal of Machine Learning Research*, 19(1):1981–2029, 2018.

R. Giordano, A. Cima, J. Murray, E. Hartman, and A. Feller. Locally equivalent weights for multilevel regression and poststratification, 2026. URL `https://arxiv.org/abs/2606.04250`.

P. Gustafson. Local sensitivity of posterior expectations. *The Annals of Statistics*, 24(1):174–195, 1996.

J. Huggins and J. Miller. Reproducible model selection using bagged posteriors. *Bayesian Analysis*, 18(1):79–104, 2023.

Feng Ji, JoonHo Lee, and Sophia Rabe-Hesketh. Valid Standard Errors for Bayesian Quantile Regression With Clustered and Independent Data. page 10769986251379738, 2025. ISSN 1076-9986. doi: 10.3102/10769986251379738. URL `https://doi.org/10.3102/10769986251379738`.

Swen Kuh, Lauren Kennedy, Qixuan Chen, and Andrew Gelman. Using leave-one-out cross validation (LOO) in a multilevel regression and poststratification (MRP) workflow: A cautionary tale. 43(5):953–982. ISSN 1097-0258. doi: 10.1002/sim.9964. URL `https://onlinelibrary.wiley.com/doi/abs/10.1002/sim.9964`.

Nanyu Luo and Feng Ji. Robust Standard Errors for Bayesian Posterior Functionals via the Infinitesimal Jackknife, 2026. URL `http://arxiv.org/abs/2604.03398`.

Krikamol Muandet, Wittawat Jitkrittum, and Jonas Kübler. Kernel Conditional Moment Test via Maximum Moment Restriction. In *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 41–50. PMLR. URL `https://proceedings.mlr.press/v124/muandet20a.html`.

Felix Otto. The Geometry of Dissipative Evolution Equations: The Porous Medium Equation. *Communications in Partial Differential Equations*, 26(1-2):101–174, January 2001. ISSN 0360-5302. doi: 10.1081/PDE-100002243. URL `https://doi.org/10.1081/PDE-100002243`. Publisher: Taylor & Francis _eprint: https://doi.org/10.1081/PDE-100002243.

Mateo Rojas-Carulla, Bernhard Scholkopf, Richard Turner, and Jonas Peters. Invariant Models for Causal Transfer Learning.

Nathan Ross. Fundamentals of Stein's method. 2011.

Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations, 2021. URL `https://arxiv.org/abs/2011.13456`.

Aki Vehtari, Andrew Gelman, and Jonah Gabry. Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. 27(5):1413–1432. ISSN 1573-1375. doi: 10.1007/s11222-016-9696-4. URL `https://doi.org/10.1007/s11222-016-9696-4`.

R. von Mises. On the asymptotic distribution of differentiable statistical functions. *The Annals of Mathematical Statistics*, 18 (3):309–348, 1947.

Yu Wang, Jie Ding, and Jonathan H. Huggins. Accurate Large-sample Uncertainty Quantification using Stochastic Gradient Markov Chain Monte Carlo, 2026. URL `http://arxiv.org/abs/2606.00293`.